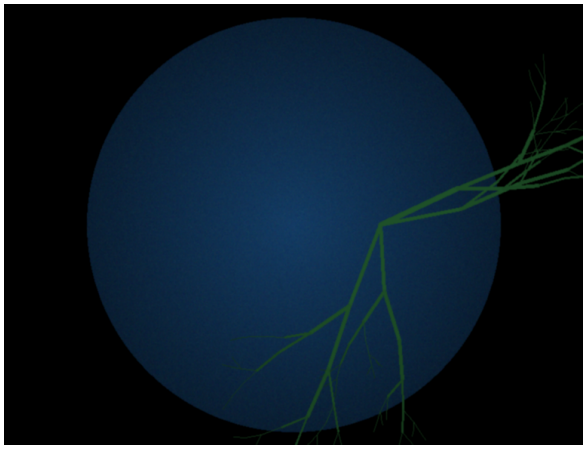
RETINAL IMAGE ANALYSIS REPORT

Image ID: **001_training** | Generated: 2026-10-21 23:31

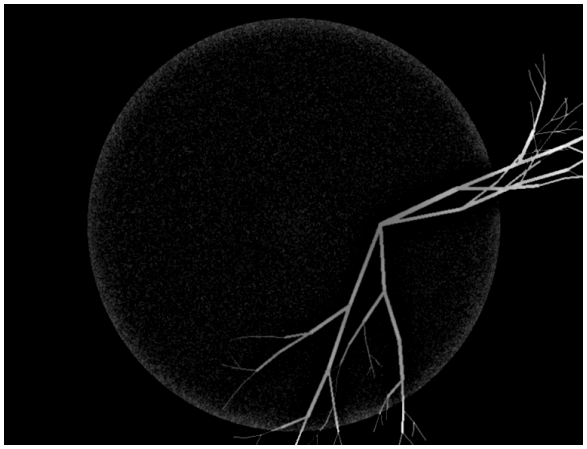
■ RESEARCH / DECISION-SUPPORT PROTOTYPE — NOT A MEDICAL DIAGNOSIS. Candidate regions detected by classical DIP algorithms. Clinical review by a qualified ophthalmologist is required.

Image dimensions: **768 × 584 pixels**

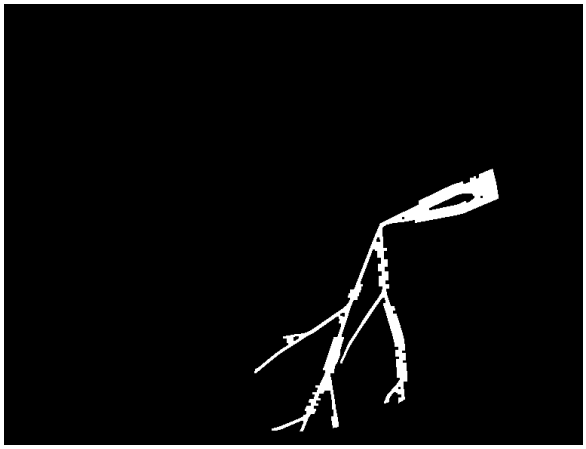
1. Pipeline Images



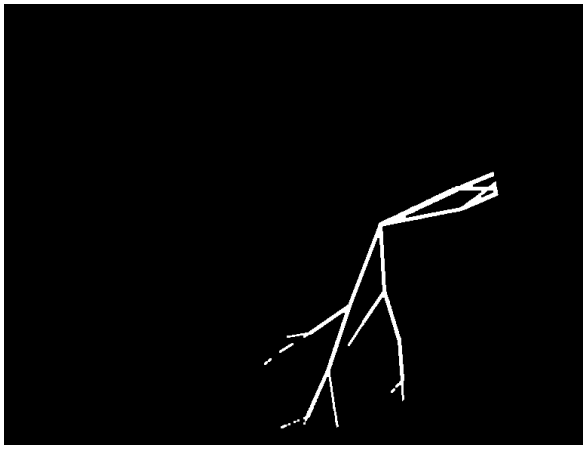
Original Fundus Image



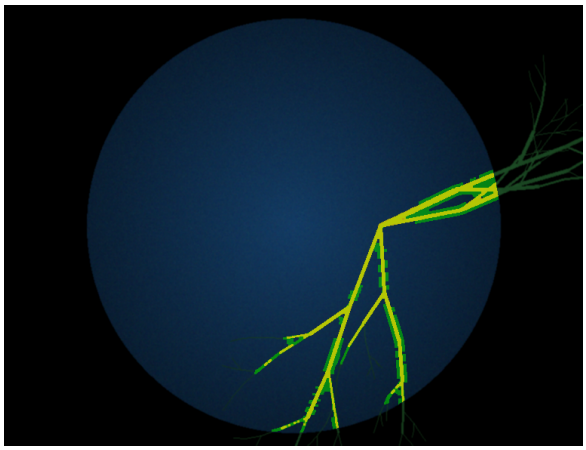
Preprocessed (Illumination-Corrected)



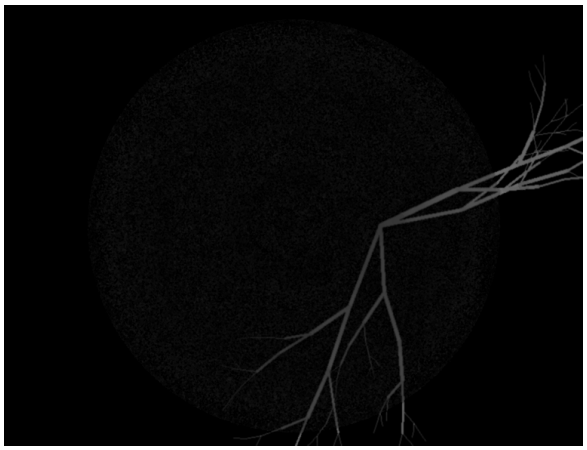
Blood-Vessel Mask



Candidate Exudate Mask



Combined Overlay



Top-Hat Response

2. Quantitative Measurements

Metric	Value
Vessel density (% of FOV)	4.56%
Candidate regions detected	6
Total candidate area (px)	6163
Largest candidate area (px)	5969

3. Candidate Region Table

ID	Area (px)	Centroid (x, y)	Bounding Box
1	5969	(515.3, 360.7)	[375, 222, 281×339]
2	73	(373.9, 454.9)	[365, 449, 19×13]
3	38	(349.5, 473.0)	[345, 469, 10×9]
4	24	(515.5, 511.5)	[513, 509, 6×6]
5	20	(387.5, 552.0)	[384, 550, 8×5]
6	39	(372.5, 558.2)	[367, 555, 13×7]

4. Evaluation Metrics

Metric	Value	Description
Dice	0.6911	2·TP / (2·TP + FP + FN)
IoU	0.5280	TP / (TP + FP + FN)
Sensitivity	0.9035	TP / (TP + FN) — recall
Specificity	0.9893	TN / (TN + FP)
Precision	0.5595	TP / (TP + FP)
F1	0.6911	Harmonic mean of precision & recall

5. Limitations

• This system uses classical DIP algorithms; it is not trained or validated as a clinical tool.

• Top-hat morphology detects bright regions, not semantic lesion categories. False positives (e.g., optic disc, image artefacts) are expected.

• Results are sensitive to image quality, acquisition conditions, and algorithm parameter choices.

• Gabor-based vessel segmentation may miss fine capillaries or produce false positives in textured regions.

• Ground-truth evaluation is only available for standard dataset images.

• This report does not constitute a medical diagnosis.